This is sample data provided for CALIBER purposes only
SDK Polyolefin Expansion Project

INTERLOCK LOGIC DIAGRAM & CAUSE / EFFECT MATRIX

DOC NO: TJC-LLD-IL-GA-1201A
WORK NO: BA-0256 REV: 3

LOGIC No: SEQ-1201 | DESCRIPTION: **HEXANE FEED PUMP (GA-1201A/B) SHUTDOWN LOGIC** | **SIL: SIL 1** | TAG: **GA-1201A** FLOC: TJC-LLD-1200-01

CAUSE & EFFECT MATRIX

ID	INITIATOR / CAUSE (INPUT)	TAG	SET POINT	VOTE	EFF-1 TRIP PUMP MOTOR GA-1201A	EFF-2 CLOSE DISCHARGE XV-1201	EFF-3 OPEN MIN-FLOW FV-1201	EFF-4 ANNUNCIATE ALARM ON DCS	EFF-5 START STANDBY PUMP GA-1201B (auto)
T1	Suction pressure LOW-LOW	**PSLL-1201**	< 0.5 barg	2oo3	X	X		X	X
T2	Discharge flow LOW-LOW (min flow)	**FSLL-1201**	< 9 m3/h for 30 s	1oo1	X		X	X	X
T3	Bearing vibration HIGH-HIGH	**VSHH-1201**	> 7.1 mm/s RMS	1oo2	X	X		X	X
T4	Bearing temperature HIGH-HIGH	**TSHH-1201**	> 95 degC	1oo1	X	X		X	
T5	Motor overload / electrical fault	**MPR-1201**	MPR relay pickup	1oo1	X	X		X	X
T6	Local/DCS emergency stop	**HS-1201**	Manual	1oo1	X	X		X	

Legend: **X** = initiator (row) actuates that effect (column). Voting: 1oo1/1oo2/2oo3 = number of transmitters that must agree to trip.

EFFECT ID	FINAL ELEMENT / ACTION
EFF-1	TRIP PUMP MOTOR GA-1201A
EFF-2	CLOSE DISCHARGE XV-1201
EFF-3	OPEN MIN-FLOW FV-1201
EFF-4	ANNUNCIATE ALARM ON DCS
EFF-5	START STANDBY PUMP GA-1201B (auto)

#	START PERMISSIVE (AND-gate)	SIGNAL
1	Suction valve OPEN	ZSO-1201
2	Seal flush established (dP > 1.5 bar)	PDI-1201
3	Min-flow valve OPEN	ZSO-1202
4	No active trip / reset done	DCS reset
=>	**ALL permissives TRUE (AND) => START GA-1201A ENABLED**	RUN

NOTES: 1. Trip set points are DUMMY training values. 2. On any trip the effects marked X are actuated by the Safety PLC and annunciated on the DCS. 3. A trip is latched and requires a manual reset once the cause has cleared and permissives are healthy. 4. Refer to P&ID; TJC-LLD-PID-1201 for instrument loops and the Datasheet for design limits of GA-1201A.

This is sample data provided for CALIBER purposes only

SDK Polyolefin Expansion Project

INTERLOCK LOGIC DIAGRAM & CAUSE / EFFECT MATRIX

DOC NO: TJC-LLD-IL-YD-2301
WORK NO: BA-0256 REV: 3

LOGIC No: SEQ-5500 | DESCRIPTION: **DRYER SYSTEM (YD-2301) SHUTDOWN LOGIC** | **SIL: SIL 2** | TAG: **YD-2301** FLOC: TJC-LLD-2300-01

CAUSE & EFFECT MATRIX

ID	INITIATOR / CAUSE (INPUT)	TAG	SET POINT	VOTE	EFF-1 TRIP DRUM DRIVE MOTOR	EFF-2 CLOSE STEAM INLET TV-2301	EFF-3 TRIP CENTRIFUGE FEED GF-2210	EFF-4 OPEN N2 PURGE FV-2302 (max inert)	EFF-5 ANNUNCIATE ALARM ON DCS
T1	Outlet temperature HIGH-HIGH	**TSHH-2301**	> 125 degC	1oo1	X	X			X
T2	Nitrogen purge flow LOW-LOW	**FSLL-2302**	< 200 kg/h	2oo3	X	X		X	X
T3	Inlet chute level HIGH-HIGH	**LSHH-2303**	> 90 %	1oo1			X		X
T4	Drum speed LOW-LOW (chain slip)	**SSLL-2305**	< 1 rpm	1oo1		X			X
T5	Vent O2 HIGH-HIGH	**ASHH-2307**	> 8 % O2	1oo2	X	X		X	X
T6	Drive motor overload	**MPR-2301**	MPR pickup	1oo1	X	X			X

Legend: **X** = initiator (row) actuates that effect (column). Voting: 1oo1/1oo2/2oo3 = number of transmitters that must agree to trip.

EFFECT ID	FINAL ELEMENT / ACTION
EFF-1	TRIP DRUM DRIVE MOTOR
EFF-2	CLOSE STEAM INLET TV-2301
EFF-3	TRIP CENTRIFUGE FEED GF-2210
EFF-4	OPEN N2 PURGE FV-2302 (max inert)
EFF-5	ANNUNCIATE ALARM ON DCS

#	START PERMISSIVE (AND-gate)	SIGNAL
1	Nitrogen inerting established (O2 < 5%)	AT-2307
2	Steam condensate drain open	ZSO-2304
3	Lube oil to gearbox OK	PSL-2306
4	No active trip / reset done	DCS reset
=>	**ALL permissives TRUE (AND) => START YD-2301 ENABLED**	RUN

Notes: 1. Trip set points are DUMMY training values. 2. On any trip the effects marked X are actuated by the Safety PLC and annunciated on the DCS. 3. A trip is latched and requires a manual reset once the cause has cleared and permissives are healthy. 4. Refer to P&ID; TJC-LLD-PID-2301 for instrument loops and the Datasheet for design limits of YD-2301.

This is sample data provided for CALIBER purposes only
SDK Polyolefin Expansion Project

INTERLOCK LOGIC DIAGRAM & CAUSE / EFFECT MATRIX

DOC NO: TJC-LLD-IL-DC-3401A
WORK NO: BA-0256 REV: 3

LOGIC No: SEQ-3401 | DESCRIPTION: **CATALYST REDUCTION REACTOR (DC-3401A) + HEATER EA-3401 LOGIC** | **SIL: SIL 2** | TAG: **DC-3401A** FLOC: TJC-LLD-3400-01

CAUSE & EFFECT MATRIX

ID	INITIATOR / CAUSE (INPUT)	TAG	SET POINT	VOTE	EFF-1 TRIP HEATER EA-3401	EFF-2 CLOSE H2 INJECTION FV-3403	EFF-3 OPEN REACTOR VENT PV-3404	EFF-4 MAINTAIN / INCREASE N2 FV-17343	EFF-5 ANNUNCIATE ALARM ON DCS
T1	Bed temperature HIGH-HIGH	**TSHH-3401**	> 230 degC	2oo3	X	X		X	X
T2	Reactor pressure HIGH-HIGH	**PSHH-3404**	> 5 barg	1oo2		X	X		X
T3	Outlet O2 HIGH during H2 (inerting fail)	**AI-3401**	> 100 ppm	1oo1		X		X	X
T4	Nitrogen carrier flow LOW-LOW	**FSLL-17343**	< 150 kg/h	1oo1	X	X			X
T5	Heater element over-temperature	**EHT-3401**	skin TC > 260 degC	1oo1	X				X
T6	Emergency stop (manual)	**HS-3401**	Manual	1oo1	X	X	X		X

Legend: **X** = initiator (row) actuates that effect (column). Voting: 1oo1/1oo2/2oo3 = number of transmitters that must agree to trip.

EFFECT ID	FINAL ELEMENT / ACTION
EFF-1	TRIP HEATER EA-3401
EFF-2	CLOSE H2 INJECTION FV-3403
EFF-3	OPEN REACTOR VENT PV-3404
EFF-4	MAINTAIN / INCREASE N2 FV-17343
EFF-5	ANNUNCIATE ALARM ON DCS

#	START PERMISSIVE (AND-gate)	SIGNAL
1	O2 free / inerting done (AI-3401 < 100 ppm)	AI-3401
2	N2 carrier flow established (> 150 kg/h)	FT-17343
3	Bed homogeneous 120 degC (all TE within 15 degC)	TE-3401
4	No active trip / reset done	DCS reset
=>	**ALL permissives TRUE (AND) => START DC-3401A ENABLED**	RUN

NOTES: 1. Trip set points are DUMMY training values. 2. On any trip the effects marked X are actuated by the Safety PLC and annunciated on the DCS. 3. A trip is latched and requires a manual reset once the cause has cleared and permissives are healthy. 4. Refer to P&ID TJC-LLD-PID-3401 for instrument loops and the Datasheet for design limits of DC-3401A.

SDK Polyolefin Expansion Project

INTERLOCK LOGIC DIAGRAM & CAUSE / EFFECT MATRIX

DOC NO: TJC-LLD-IL-KC-4501
WORK NO: BA-0256 REV: 3

LOGIC No: SEQ-4501 | DESCRIPTION: **RECYCLE GAS COMPRESSOR (KC-4501) SHUTDOWN LOGIC** | **SIL: SIL 2** | TAG: **KC-4501** FLOC: TJC-LLD-4500-01

CAUSE & EFFECT MATRIX

ID	INITIATOR / CAUSE (INPUT)	TAG	SET POINT	VOTE	EFF-1 TRIP COMPRESSOR MOTOR KC-4501	EFF-2 OPEN ANTI-SURGE RECYCLE FV-4502	EFF-3 CLOSE SUCTION XV-4501	EFF-4 MAINTAIN SEAL GAS N2	EFF-5 ANNUNCIATE ALARM ON DCS
T1	Lube oil pressure LOW-LOW	**PSLL-4504**	< 1.5 barg	2oo3	X	X	X		X
T2	Discharge temperature HIGH-HIGH	**TSHH-4503**	> 140 degC	1oo1	X	X			X
T3	Discharge pressure HIGH-HIGH	**PSHH-4502**	> 14 barg	1oo2	X	X			X
T4	Crosshead vibration HIGH-HIGH	**VSHH-4505**	> 11 mm/s	1oo2	X	X	X		X
T5	Suction KO drum level HIGH-HIGH	**LSHH-4510**	> 80 %	1oo1	X		X		X
T6	Suction pressure LOW-LOW	**PSLL-4501**	< 0.3 barg	1oo1	X	X			X

Legend: **X** = initiator (row) actuates that effect (column). Voting: 1oo1/1oo2/2oo3 = number of transmitters that must agree to trip.

EFFECT ID	FINAL ELEMENT / ACTION
EFF-1	TRIP COMPRESSOR MOTOR KC-4501
EFF-2	OPEN ANTI-SURGE RECYCLE FV-4502
EFF-3	CLOSE SUCTION XV-4501
EFF-4	MAINTAIN SEAL GAS N2
EFF-5	ANNUNCIATE ALARM ON DCS

#	START PERMISSIVE (AND-gate)	SIGNAL
1	Lube oil pressure OK (> 2 barg)	PSLL-4504
2	Seal gas established	FT-4506
3	KO drum level normal (< 50%)	LT-4510
4	Jacket water flow OK	FSL-4508
5	No active trip / reset done	DCS reset
=>	**ALL permissives TRUE (AND) => START KC-4501 ENABLED**	RUN

NOTES: 1. Trip set points are DUMMY training values. 2. On any trip the effects marked X are actuated by the Safety PLC and annunciated on the DCS. 3. A trip is latched and requires a manual reset once the cause has cleared and permissives are healthy. 4. Refer to P&ID TJC-LLD-PID-4501 for instrument loops and the Datasheet for design limits of KC-4501.

This is sample data provided for
CALIBER purposes only
SDK Polyolefin Expansion Project

INTERLOCK LOGIC DIAGRAM & CAUSE / EFFECT MATRIX

DOC NO: TJC-LLD-IL-EA-5601
WORK NO: BA-0256 REV: 3

LOGIC No: N/A (control loop only)	DESCRIPTION: **SOLVENT HEATER EA-5601 - TEMPERATURE CONTROL (no dedicated ESD trip)**	**SIL: N/A**	TAG: **EA-5601** FLOC: TJC-LLD-5600-01

CAUSE & EFFECT MATRIX

ID	INITIATOR / CAUSE (INPUT)	TAG	SET POINT	VOTE	EFF-1 MODULATE STEAM VALVE TV-5602	EFF-2 ANNUNCIATE HIGH dP / FOULING ALARM	EFF-3 MECHANICAL RELIEF VIA PSV-5607
C1	Outlet temperature control	**TIC-5602**	SP 95 degC	control	X		
A1	Tube-side dP HIGH (fouling)	**PDAH-5605**	> 0.7 bar	alarm		X	
R1	Tube-side overpressure relief	**PSV-5607**	set 16 barg	mech			X

Legend: **X** = initiator (row) actuates that effect (column). Voting: 1oo1/1oo2/2oo3 = number of transmitters that must agree to trip.

EFFECT ID	FINAL ELEMENT / ACTION
EFF-1	MODULATE STEAM VALVE TV-5602
EFF-2	ANNUNCIATE HIGH dP / FOULING ALARM
EFF-3	MECHANICAL RELIEF VIA PSV-5607

#	START PERMISSIVE (AND-gate)	SIGNAL
1	Solvent flow established (from GA-5610)	FSL upstream
2	Condensate trap draining (TI-5604 < steam T)	TI-5604
=>	**ALL permissives TRUE (AND) => START EA-5601 ENABLED**	RUN

NOTES: 1. Trip set points are DUMMY training values. 2. On any trip the effects marked X are actuated by the Safety PLC and annunciated on the DCS. 3. A trip is latched and requires a manual reset once the cause has cleared and permissives are healthy. 4. Refer to P&ID TJC-LLD-PID-5601 for instrument loops and the Datasheet for design limits of EA-5601.

This is sample data provided for
CALIBER purposes only
SDK Polyolefin Expansion Project

INTERLOCK LOGIC DIAGRAM & CAUSE / EFFECT MATRIX

DOC NO: TJC-LLD-IL-LV-6701
WORK NO: BA-0256 REV: 3

LOGIC No: SEQ-6701 | DESCRIPTION: **LP SEPARATOR LEVEL PROTECTION (FA-6710 / LV-6701) LOGIC** | **SIL: SIL 1** | TAG: **LV-6701** FLOC: TJC-LLD-6700-01

CAUSE & EFFECT MATRIX

ID	INITIATOR / CAUSE (INPUT)	TAG	SET POINT	VOTE	EFF-1 OPEN LV-6701 FULLY (relieve level)	EFF-2 CLOSE LV-6701 (prevent gas blow-by)	EFF-3 VENT ACTUATOR - LV-6701 FAILS CLOSED	EFF-4 ANNUNCIATE ALARM ON DCS
T1	Separator level HIGH-HIGH	**LSHH-6710**	> 85 %	1oo2	X			X
T2	Separator level LOW-LOW	**LSLL-6710**	< 15 %	1oo2		X		X
T3	ESD activation (manual)	**HS-6701**	ESD signal	1oo1			X	X
T4	Instrument air pressure LOW	**PSL-6702**	< 1.0 barg	1oo1			X	X

Legend: **X** = initiator (row) actuates that effect (column). Voting: 1oo1/1oo2/2oo3 = number of transmitters that must agree to trip.

EFFECT ID	FINAL ELEMENT / ACTION
EFF-1	OPEN LV-6701 FULLY (relieve level)
EFF-2	CLOSE LV-6701 (prevent gas blow-by)
EFF-3	VENT ACTUATOR - LV-6701 FAILS CLOSED
EFF-4	ANNUNCIATE ALARM ON DCS

#	START PERMISSIVE (AND-gate)	SIGNAL
1	Instrument air available (> 1.2 barg)	PI-6702
2	Positioner DVC6200 in AUTO/healthy	DVC6200
3	Bypass HV-6701 closed (normal operation)	HV-6701
=>	**ALL permissives TRUE (AND) => START LV-6701 ENABLED**	RUN

NOTES: 1. Trip set points are training values. 2. On any trip the effects marked X are actuated by the Safety PLC and annunciated on the DCS. 3. A trip is latched and requires a manual reset once the cause has cleared and permissives are healthy. 4. Refer to P&ID TJC-LLD-PID-6701 for instrument loops and the Datasheet for design limits of LV-6701.

This is sample data provided
for CALIBER purposes only

INTERLOCK LOGIC DIAGRAM & CAUSE / EFFECT MATRIX

DOC NO: TJC-LLD-IL-CT-7801
WORK NO: BA-0256 REV: 3

LOGIC No: SEQ-7801	DESCRIPTION: **COOLING TOWER FAN (CT-7801) SHUTDOWN LOGIC**	SIL: SIL 1	TAG: **CT-7801** FLOC: TJC-LLD-7800-01

CAUSE & EFFECT MATRIX

ID	INITIATOR / CAUSE (INPUT)	TAG	SET POINT	VOTE	EFF-1 TRIP FAN MOTOR CT-7801	EFF-2 ANNUNCIATE ALARM ON DCS	EFF-3 REQUEST START OF SPARE CELL CT-7802
T1	Fan/gearbox vibration HIGH-HIGH	**VSHH-7802**	> 9 mm/s	1oo1	X	X	X
T2	Gearbox oil temperature HIGH-HIGH	**TSHH-7803**	> 90 degC	1oo1	X	X	X
T3	Gearbox oil pressure LOW	**PSL-7807**	< 0.8 barg	1oo1	X	X	
T4	Fan motor overload	**MPR-7801**	MPR pickup	1oo1	X	X	X

Legend: **X** = initiator (row) actuates that effect (column). Voting: 1oo1/1oo2/2oo3 = number of transmitters that must agree to trip.

EFFECT ID	FINAL ELEMENT / ACTION
EFF-1	TRIP FAN MOTOR CT-7801
EFF-2	ANNUNCIATE ALARM ON DCS
EFF-3	REQUEST START OF SPARE CELL CT-7802

#	START PERMISSIVE (AND-gate)	SIGNAL
1	Gearbox oil level OK	LG gearbox
2	Basin level normal (> LSL-7804)	LT-7804
3	No fan lockout / maintenance key released	lockout
=>	**ALL permissives TRUE (AND) => START CT-7801 ENABLED**	RUN

NOTES: 1. Trip set points are DUMMY training values. 2. On any trip the effects marked X are actuated by the Safety PLC and annunciated on the DCS. 3. A trip is latched and requires a manual reset once the cause has cleared and permissives are healthy. 4. Refer to P&ID; TJC-LLD-PID-7801 for instrument loops and the Datasheet for design limits of CT-7801.

This is sample data provided
for CALIBER purposes only

INTERLOCK LOGIC DIAGRAM & CAUSE / EFFECT MATRIX

DOC NO: TJC-LLD-IL-FA-8901
WORK NO: BA-0256 REV: 3

LOGIC No: SEQ-8901	DESCRIPTION: **REFLUX ACCUMULATOR (FA-8901) OVERHEAD PROTECTION LOGIC**	SIL: SIL 1	TAG: **FA-8901** FLOC: TJC-LLD-8900-01

CAUSE & EFFECT MATRIX

ID	INITIATOR / CAUSE (INPUT)	TAG	SET POINT	VOTE	EFF-1 TRIP COLUMN FEED (prevent carryover)	EFF-2 TRIP REFLUX PUMP GA-8920 (protect)	EFF-3 OPEN VENT PCV-8905 (relieve pressure)	EFF-4 ANNUNCIATE ALARM ON DCS
T1	Drum level HIGH-HIGH	**LSHH-8901**	> 85 %	1oo2	X			X
T2	Drum level LOW-LOW	**LSLL-8901**	< 15 %	1oo2		X		X
T3	Drum pressure HIGH-HIGH	**PSHH-8902**	> 8 barg	1oo2			X	X
T4	Emergency stop (manual)	**HS-8901**	Manual	1oo1	X	X	X	X

Legend: **X** = initiator (row) actuates that effect (column). Voting: 1oo1/1oo2/2oo3 = number of transmitters that must agree to trip.

EFFECT ID	FINAL ELEMENT / ACTION
EFF-1	TRIP COLUMN FEED (prevent carryover)
EFF-2	TRIP REFLUX PUMP GA-8920 (protect)
EFF-3	OPEN VENT PCV-8905 (relieve pressure)
EFF-4	ANNUNCIATE ALARM ON DCS

#	START PERMISSIVE (AND-gate)	SIGNAL
1	Level in normal band (20-80%)	LT-8901
2	N2 blanket healthy	PT-8902
3	Boot not flooded	LT-8903
4	No active trip / reset done	DCS reset
=>	**ALL permissives TRUE (AND) => START FA-8901 ENABLED**	RUN

NOTES: 1. Trip set points are DUMMY training values. 2. On any trip the effects marked X are actuated by the Safety PLC and annunciated on the DCS. 3. A trip is latched and requires a manual reset once the cause has cleared and permissives are healthy. 4. Refer to P&ID; TJC-LLD-PID-8901 for instrument loops and the Datasheet for design limits of FA-8901.